

Mio Crilley

crilleym@umich.edu | [linkedin.com/in/miocrilley/](https://www.linkedin.com/in/miocrilley/) | [crilleym.github.io](https://github.com/crilley) | Novi, MI 48375 | 248.704.6569

Education

University of Michigan, Ann Arbor – B.S.E in Mechanical Engineering, Energy Systems Concentration *Expected: May 2028*

GPA: 3.32/4.0

Awards: Dean's List, University Honors, William J. Branstrom Freshman Prize, Blue Ribbon UROP Symposium

Work Experience

Consumers Energy

Livonia, MI

Co-op, Gas Engineering Standards Services

August 2026 - May 2027

- Develop and revise enterprise engineering standards and work specifications governing methane emissions measurement, collection, and net-zero reporting across gas operations.
- Author and update Electrical, Instrumentation & Controls (EI&C) engineering standards supporting the design, operation, and maintenance of gas infrastructure.
- Lead conversion and modernization of 20+ legacy technical documentation into internal templates.

Consumers Energy

Livonia, MI

Gas Engineering Intern, External Corrosion

May 2026 - August 2026

- Analyzed external corrosion on gas distribution & transmission pipelines utilizing Direct Assessment process on 15+ excavations.
- Managed facilities attribute data collection and historical information of 3 citygates and 2 valve sites.
- Organized and renamed over 800 Owl test stations to improve BSI database accuracy.

Adient - General Motors Business Unit

Plymouth, MI

Complete Seat Engineering Intern

May 2025 - August 2025

- Redesigned ventilation system for GM production seat, projecting \$358K annual savings and 8-second JIT assembly takt time reduction.
- Developed and executed tactical engineering plan, managing global team of product engineers.
- Applied root cause analysis using Adient's Basic Engineering Process to resolve cushion airflow performance issues.
- Conducted Complete Seat Airflow and Steady State Heat Flux tests in Thermal Comfort Lab; processed by comparing with FEA results and visualized test data to evaluate cooling effectiveness.
- Extensive experience rebuilding seats and prototyping including custom foam airflow test fixtures.

University of Michigan Undergraduate Instructor

Ann Arbor, MI

Peer Facilitator (UROP)

August 2026 - May 2027

- Instruct a cohort of 35 peers within the Undergraduate Research Opportunities Program (UROP).
- Manage continuous student support systems, utilizing 1-on-1 consultations and targeted review sessions to optimize academic and professional development.

Project Experience

MRacing Formula SAE

Ann Arbor, MI

Suspension Team

August 2024 - Current

- Designed the Chassis Insert Testing Assembly and over-wing push bar in Siemens NX, producing 20+ detailed manufacturing drawings with GD&T and tolerancing.
- Machined, built & tested hardware using a manual mill, lathe, horizontal band saw, and assisted with carbon layups.

University of Michigan Aerospace Department Undergraduate Research

Ann Arbor, MI

Fabrication & Testing Lead

August 2024 - May 2026

- Co-authored experimental-computational mechanics study on compressive collapse behavior of 3D-printed Kresling tubes.
- Designed tubular structures in Blender with origami and kirigami features to study compression-torsion coupling for energy absorption after compression tests with load frame in Composite Structures lab
- Analyzed deformation data using Digital Image Correlation (DIC) in GOM Correlate to assess structural collapse mechanisms.

Robotics Project - Manufacturing and Design I

Ann Arbor, MI

Team Lead

August 2025 - May 2026

- Designed 10+ custom CAD components for a mobile pick-and-place robot, including the object-hooking & lasso-pivot-rod mechanism, structural mounts, and mechanical interfaces, using SolidWorks.
- Machined 15+ components from raw stock using a manual mill, lathe, and other shop equipment, applying GD&T, dimensional tolerances, and design-for-manufacturability principles.
- Designed and fabricated a hook-arm mechanism capable of reliably engaging and lifting objects up to 1 lb, iterating component geometry and clearances based on physical testing.

Skills

CAD & Design: Siemens NX, Solidworks, CATIA, Blender, GD&T

Programming & Analysis: MATLAB, C++, structural testing, experimental data analysis

Fabrication & Testing: Manual mill, lathe, horizontal band saw, carbon layup, prototyping, mechanical testing

Productivity: Microsoft Office (Excel, PowerPoint, Word)

Languages: English (native), Japanese (proficient)